

Customer Behavior Analysis Report:-

1. Introduction

This project focuses on analyzing customer transactions and behavior to identify customer segments, purchasing patterns, and churn risks. The analysis helps Alfido Tech understand customer value and improve engagement strategies.

2. Dataset Description

The dataset contains customer transaction details, including customer information, purchase history, product details, payment methods, and purchase amounts.

3. Data Cleaning & Feature Engineering

The dataset was cleaned by:

- Checking and handling missing values
- Removing duplicate records
- Correcting data types
- Converting purchase dates into proper date format

New features were created to improve analysis, such as:

- Age groups
- Purchase trends based on dates
- Customer value metrics

4. Customer Segmentation (RFM Analysis)

RFM analysis was performed based on:

- **Recency:** How recently a customer purchased
- **Frequency:** How often a customer purchases
- **Monetary:** Total customer spending

Customers were divided into four segments:

Champions

High-value customers with frequent purchases and high spending.

Loyal Customers

Regular customers who consistently interact with the business.

Potential Loyalists

Customers with growth potential who can become loyal customers.

At Risk Customers

Customers showing reduced engagement and possible churn risk.

5. Key Findings

- Champions contribute significantly to overall revenue.
- Loyal customers provide stable business value.
- Potential loyalists can be converted through targeted marketing.
- At-risk customers require retention strategies to prevent churn.
- Customer behavior analysis helps in making data-driven decisions.

6. Recommendations for Alfido Tech

1. **Reward High-Value Customers**

Introduce loyalty programs and exclusive offers for Champions.

2. **Re-engage At-Risk Customers**

Use personalized discounts and reminder campaigns to bring inactive customers back.

3. **Improve Customer Personalization**

Provide product recommendations based on customer preferences.

4. **Increase Customer Retention**

Create membership benefits to encourage repeat purchases.

5. **Monitor Customer Segments Regularly**

Use customer analytics to track behavior changes and reduce churn.

7. Conclusion

The analysis provides valuable insights into customer purchasing behavior. By focusing on customer retention, personalization, and targeted engagement strategies, Alfido Tech can improve customer satisfaction and business growth.

Submitted by: Tapaswini Shaw

Candidate ID: IS-2026-11564

Course: B.Tech CSE (Artificial Intelligence)

Internship: InternshipSpark – Data Analyst